CIS 353 Project Proposal

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Idea & Introduction

Gym Performance Tracking System - Most current gyms rely on data-driven insights to help their customers achieve their fitness goals. Our domain application will be implemented with the intent of storing, managing, and analyzing information related to any "clients", how they conduct workouts, their overall measurements, and general progress over time. The database in question will support both trainers and members by providing a centralized platform to analyze fitness metrics such as BMI, weight, heart rate, and any other applicable metric.

This system will capture both static information (Name, Demographics, etc.) and dynamic information (workout logs, body measurements, etc.). In this way, the created database will complement more advanced querying results, predicting client behavior and implementing more efficient workouts for every client's needs. Our goal is to create a system that supports general operational needs and provides insightful information at the same time.

Database Requirements

***Entities in Bold**

Member: The Member entity represents people who have signed up for a membership to the gym. Each member will have attributes such as MemberID, Name, Age, Gender, Contact Information, and Membership Type. The member entity is the core of the database, which allows connections to all of the other entities within our system. We can also use additional attributes like membership status (whether or not a membership is active or not active) and enrollment date to more accurately track progress over time.

Member Phone: This entity represents multiple phone numbers associated with a single member. Since a member can have more than one phone number, this entity is a multi-valued attribute of the Member entity. Each entry links a MemberID to a specific phone number.

Membership Account: This entity represents the subscription details of gym members. It has the type of membership, start/end dates, monthly fee and payment status. This allows tracking the history of the membership and is uniquely identified by AccountID.

Measurement: The Measurement entity stores physical health metrics recorded periodically for each member. Attributes include MeasurementID, Date, Weight, Height, Body Fat Percentage, and Muscle Mass. It enables tracking of progress and supports analytical queries such as average weight change or BMI trends.

Measurement Note: This entity is a weak entity that stores notes related to specific measurement records. It depends on the Measurement entity for identification and can be identified by both MeasurementID and NoteID. This allows multiple notes to be recorded for a single measurement.

Workout Session: The Workout Session entity records every event of a member coming to the gym or doing a workout. Attributes include SessionID, Date, Duration, Calories Burned, and Workout Type (e.g., cardio, strength training). These sessions are always attached to a specific member and can be guided/supervised by a trainer. This entity helps to improve workout efficiency and can lead to better physical performance.

Exercise: The Exercise entity is indicative of the specific exercises that any client can go through (though this does not represent ALL exercises). Attributes include ExerciseID, Name, Category (e.g., strength, cardio), Muscle Group Targeted, and Equipment Required. Since exercises are finite and can target the same body parts, they can be reused in a workout, making a many-to-many relationship.

Exercise Muscle Group: This entity represents the multiple muscle groups targeted by a specific exercise. Since an exercise can target more than one muscle group, this entity is a multi-valued attribute of Exercise. Each record associates an exercise with a muscle group, enabling more detailed analysis of workouts.

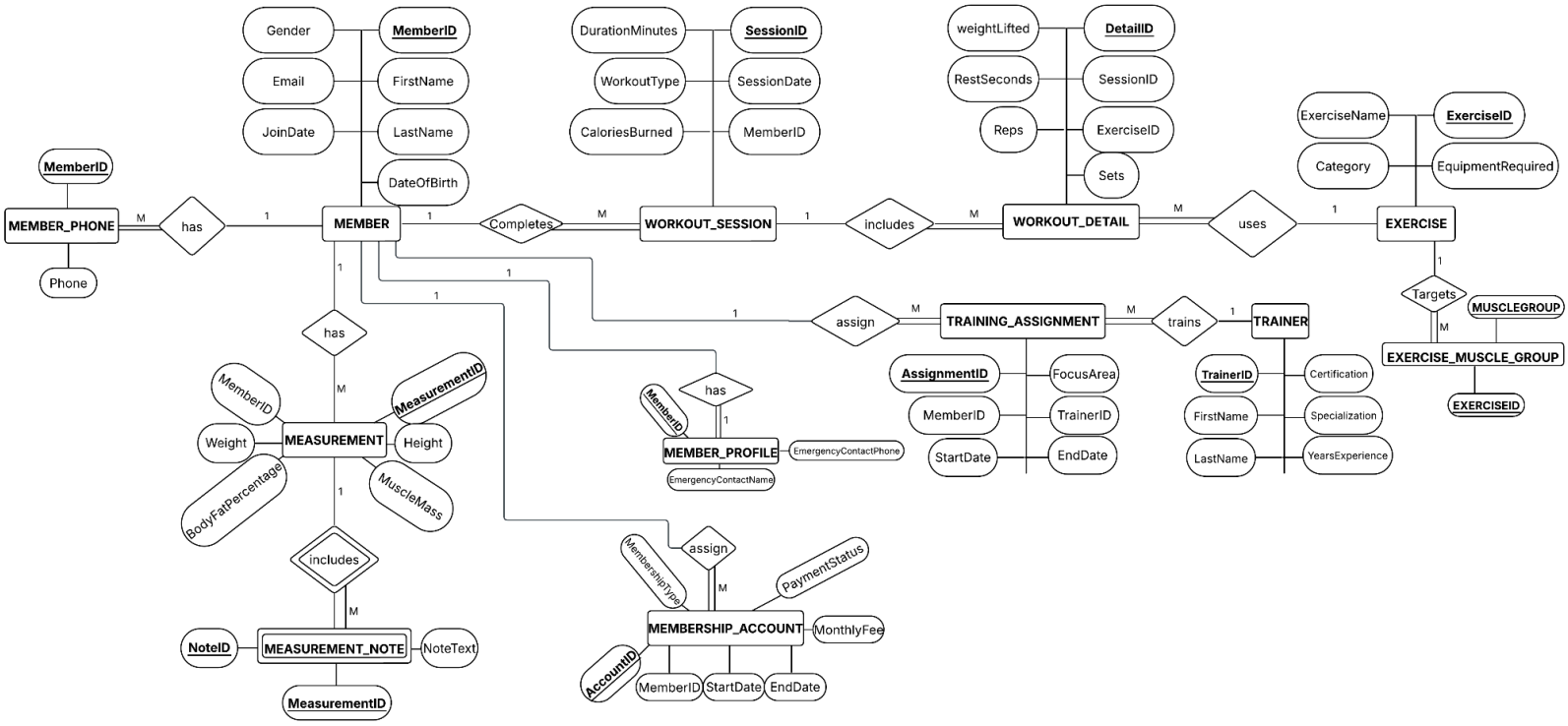
Trainer: The Trainer entity represents employees who are responsible (in some shape or form) for training members. Attributes include TrainerID, Name, Certification, Specialization, and Years of Experience. Since a trainer can train multiple members and vice-versa, this forms a many-to-many relationship.

Training Assignment: This entity represents the relationship between members and trainers. It is a many-to-many relationship, allowing a trainer to work with multiple members and a member to have multiple trainers over time. Attributes such as start date, end date, and focus area describe the nature of each assignment.

Workout Detail: The Workout Detail entity is an associative entity that represents exercises performed during a specific workout session. It has a many-to-many relationship between Workout_Session and Exercise. Attributes include DetailID, sessionID, ExerciseID, Set Number, Repetitions, Weight Lifted, and Rest Time for each exercise in a session.

Member Profile: This entity stores additional one-to-one information about a member, specifically their emergency contact information. Each profile is uniquely associated with one member and cannot exist independently.

ER Diagram



Rudimentary Relational Schema

MEMBER(**MemberID**, FirstName, LastName, DateOfBirth, Gender, Email, JoinDate, MembershipStatus)

MEMBER_PHONE(MemberID, Phone)

MEMBERSHIP_ACCOUNT(**AccountID**, MemberID, MembershipType, StartDate, EndDate, MonthlyFee, PaymentStatus)

MEASUREMENT(**MeasurementID**, MemberID, MeasureDate, Weight, Height, BodyFatPercent, MuscleMass)

WORKOUT_SESSION(**SessionID**, MemberID, SessionDate, DurationMinutes, CaloriesBurned, WorkoutType)

EXERCISE(**ExerciseID**, ExerciseName, Category, EquipmentRequired)

EXERCISE_MUSCLE_GROUP(ExerciseID, MuscleGroup)

TRAINER(TrainerID, FirstName, LastName, Certification, Specialization, YearsExperience)

TRAINING_ASSIGNMENT(AssignmentID, MemberID, TrainerID, StartDate, EndDate, FocusArea)

WORKOUT_DETAIL(DetailID, SessionID, ExerciseID, Sets, Reps, WeightLifted, RestSeconds)

MEASUREMENT_NOTE(MeasurementID, NoteID, NoteText**)**

MEMBER_PROFILE(MemberID, EmergencyContactName**,** EmergencyContactPhone**)**

Integrity Constraints

Constraint Name & Table	Constraint Type	English Description
MembershipAccount_Member_FK	Foreign Key	Ensures that every membership account belongs to a valid member in the MEMBER table.
Measurement_Member_FK	Foreign Key	Ensures that every measurement record is associated with an existing member.
WorkoutSession_Member_FK	Foreign Key	Ensures that every workout session is linked to a valid member.
Member_PK	Primary Key	MemberID uniquely Identifies each member
MemberPhone_PK	Primary Key	(MemberID, Phone) uniquely identifies each phone number for a member
MemberPhone_Member_FK	Foreign Key	Each phone belongs to a valid member
MembershipAccount_PK	Primary Key	AccountID identifies each membership account
Measurement_PK	Primary Key	MeasurementID identifies each workout sesh

WorkoutSession_PK	Primary Key	SessionID identifies each workout session
Exercise_PK	Primary Key	ExerciseID identifies each exercise
ExerciseMuscleGroup_PK	Primary Key	(ExerciseID, MuscleGroup) identifies each muscle group per exercise
ExerciseMuscleGroup_Exercise_FK	Foreign Key	Ensures each muscle group entry references a valid exercise
Trainer_PK	Primary Key	TrainerID identifies each trainer
TrainingAssignment_PK	Primary Key	AssignmentID identifies each training assign
WorkoutDetail_PK	Primary Key	(SessionID, ExerciseID) identifies each exercise within a session
MeasurementNote_PK	Primary Key	(MeasurementID, NoteID) identifies each note
MeasurementNote_Measurement_FK	Foreign Key	Ensures each note belongs to a valid measurement
MemberProfile_PK	Primary Key	MemberID identifies each member profile
TrainingAssignment_Member_FK	Foreign Key	Ensures that every training assignment references a valid member.
TrainingAssignment_Trainer_FK	Foreign Key	Ensures that every training assignment references a valid trainer.
WorkoutDetail_Session_FK	Foreign Key	Ensures that every workout detail belongs to an existing workout session.
WorkoutDetail_Exercise_FK	Foreign Key	Ensures that every workout detail references a valid exercise.
Member_Gender_CC	1-attribute	Restricts Gender to allowed values such as ‘Male’, ‘Female’, or ‘Other’.
Member_Status_CC	1-attribute	Restricts MembershipStatus to allowed values such as ‘Active’, ‘Inactive’, or ‘Suspended’.
MembershipAccount_Type_CC	1-attribute	Restricts MembershipType to valid categories such as ‘Basic’, ‘Premium’, or ‘Platinum’
MembershipAccount_Fee_CC	1-attribute	Ensures that MonthlyFee is greater than or equal to 0.

MemberProfile_Member_FK	Foreign Key	Each profile belongs to a valid member
Measurement_BodyFat_CC	1-attribute	Ensures that BodyFatPercent is between 0 and 100.
Measurement_MuscleMass_CC	1-attribute	Ensures that MuscleMass is greater than or equal to 0.
WorkoutSession_Duration_CC	1-attribute	Ensures that DurationMinutes is greater than 0.
WorkoutSession_Calories_CC	1-attribute	Ensures that CaloriesBurned is greater than or equal to 0.
Exercise_Category_CC	1-attribute	Restricts Category to valid values such as ‘Strength’, ‘Cardio’, ‘Flexibility’, or ‘Endurance’.
Trainer_Experience_CC	1-attribute	Ensures that YearsExperience is greater than or equal to 0.
WorkoutDetail_Sets_CC	1-attribute	Ensures that Sets is greater than 0.
WorkoutDetail_Reps_CC	1-attribute	Ensures that Reps is greater than 0.
WorkoutDetail_WeightLifted_CC	1-attribute	Ensures that WeightLifted is greater than or equal to 0.
WorkoutDetail_RestSeconds_CC	1-attribute	Ensures that RestSeconds is greater than or equal to 0.
MembershipAccount_DateRange_CC	2-attribute, 1-row	Ensures that EndDate is later than or equal to StartDate for a membership account.
TrainingAssignment_DateRange_CC	2-attribute, 1-row	Ensures that EndDate is later than or equal to StartDate for a trainer assignment.
Measurement_HeightWeight_CC	2-attribute, 1-row	Ensures that Height and Weight are both positive whenever a measurement record is stored.
WorkoutDetail_SetsReps_CC	2-attribute, 1-row	Ensures that if Sets is recorded, both sets and reps are greater than 0

https://www.kaggle.com/datasets/valakhorasani/gym-members-exercise-dataset
https://www.kaggle.com/datasets/ambarishdeb/gym-exercises-dataset